

TRANSFER LEARNING FOR MULTI-LABEL RETINAL DISEASE CLASSIFICATION

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ABSTRACT

This report presents a comprehensive study on transfer learning for multi-label retinal disease classification using the ODIR dataset. We investigate transfer learning strategies, class imbalance mitigation, attention mechanisms, and ensemble methods to improve detection of Diabetic Retinopathy, Glaucoma, and Age-related Macular Degeneration. Our best ensemble model (simple/weighted averaging) achieves an average F1-score of **84.476%** on the onsite test set, representing a 1.78% improvement over the best single model and a 49.0% relative improvement over the no fine-tuning baseline.

1. INTRODUCTION

Medical image classification faces challenges including limited annotated data and severe class imbalance. Transfer learning addresses these by leveraging pretrained models. We focus on multi-label classification of retinal diseases using the ODIR dataset (800 training, 200 validation, 300 offsite test, 250 onsite test images at 256×256 resolution).

The dataset exhibits significant class imbalance: DR (~600 images), Glaucoma (~150), and AMD (~100). Our contributions include: (1) comparative analysis of transfer learning strategies, (2) specialized loss functions for imbalanced data, (3) attention mechanisms, and (4) ensemble methods with explainability analysis.

2. METHODS

2.1. Model Architecture and Transfer Learning

We employ EfficientNet-B0 [1] and ResNet18 [2], pretrained on ADAM, REFUGE2, and APTOS datasets. We evaluate three strategies: (1) **No fine-tuning** - direct evaluation, (2) **Classifier fine-tuning** - frozen backbone with trainable classifier (lr=1e-3), and (3) **Full fine-tuning** - end-to-end training with lr=1e-4.

2.2. Class Imbalance Mitigation

Focal Loss: Down-weights easy examples via $FL(p_t) = -\alpha_t(1 - p_t)^\gamma \log(p_t)$ with $\alpha = 0.25$, $\gamma = 2.0$ [3].

Class-Balanced Loss: Re-weights BCE using $w_j = \frac{1-\beta}{1-\beta^{n_j}}$ where n_j is sample count and $\beta = 0.9999$ [4].

2.3. Attention Mechanisms

SE Blocks: Channel-wise recalibration through squeeze (global pooling) and excitation (FC layers) operations [5].

Multi-Head Attention: Self-attention with 8 heads computing $\text{Attention}(Q, K, V) = \text{softmax}(QK^T/\sqrt{d_k})V$ [6].

2.4. Ensemble Methods

The ensemble combines three ResNet18 models: (1) Full fine-tuning (Task 1-3), (2) Focal Loss (Task 2-1), and (3) Class-Balanced Loss (Task 2-2), selected for their complementary strengths across disease classes.

Simple Average: Equal-weight averaging: $p_{final} = \frac{1}{N} \sum_{i=1}^N p_i$.

Weighted Average: Performance-based weighting using validation F1-scores: $p_{final} = \sum_{i=1}^N w_i p_i$ where $w_i = F1_i / \sum F1_j$.

Max Voting: Binary predictions combined via majority vote.

Stacking: Meta-learner (2-layer MLP, 32 hidden units) trained on validation set to optimally combine base model predictions.

2.5. Explainability Analysis

We employ GradCAM [7] to visualize model attention, computing class-discriminative localization maps: $L_{GradCAM} = ReLU(\sum_k w_k A_k)$ where w_k are gradient-derived weights and A_k are feature maps.

2.6. Training Configuration

Data augmentation includes random flips, rotation ($\pm 15^\circ$), and color jittering. Adam optimizer with batch size 32, training for 30 epochs. Learning rate reduced by 50% on validation plateau (patience=3). Best model selected by validation F1-score.

3. RESULTS

Complete results are in Appendix A.

3.1. Transfer Learning (Task 1)

Full fine-tuning achieves best performance (78.99%), showing 27.73% relative improvement over no fine-tuning baseline. ResNet18 outperforms EfficientNet by 7.03 percentage

Table 1: Transfer Learning - Offsite F1-score (%)

Method	No FT	Classifier	Full FT
EfficientNet	55.41	70.88	71.96
ResNet18	51.26	56.80	78.99

points, likely due to ResNet’s residual connections facilitating gradient flow in the limited-data regime, while EfficientNet’s larger capacity leads to overfitting on 800 training images.

The onsite results can be found in Appendix B (Task 1-4).

3.2. Loss Functions (Task 2)

Table 2: Loss Function Comparison - ResNet18 - F1-score (%)

Loss	DR	G	AMD	Avg
BCE (offsite)	92.31	73.56	71.11	78.99
Focal (offsite)	88.97	74.16	62.50	75.21
CB (offsite)	86.26	78.43	62.50	75.73
BCE (onsite)	-	-	-	81.84
Focal (onsite)	-	-	-	83.00
CB (onsite)	-	-	-	82.45

An interesting observation: specialized loss functions show minimal improvement on offsite test but significantly improve onsite performance (Focal: +1.16%, CB: +0.61% over BCE). This suggests these techniques generalize better to unseen data despite lower validation scores, highlighting the importance of held-out test evaluation.

3.3. Attention Mechanisms (Task 3)

Table 3: Attention Results - ResNet18 - F1-score (%)

Method	DR	G	AMD	Avg
No Attention	92.31	73.56	71.11	78.99
SE Block	89.22	79.12	65.38	77.91
MHA	83.46	80.00	51.72	71.73

Attention mechanisms do not improve offsite results but enhance onsite performance (SE: 82.31%, MHA: 80.60% vs BCE: 81.84%), with SE Block significantly outperforming MHA, likely due to MHA overfitting on the small dataset.

3.4. Ensemble Methods (Task 4)

Simple and weighted averaging achieve identical performance (84.48%), as optimal weights [0.495, 0.505] are nearly equal. This represents a 1.78% improvement over the best single model (Focal Loss: 83.00%). Each model contributes complementary strengths: BCE excels on DR, Focal/CB improve minority classes.

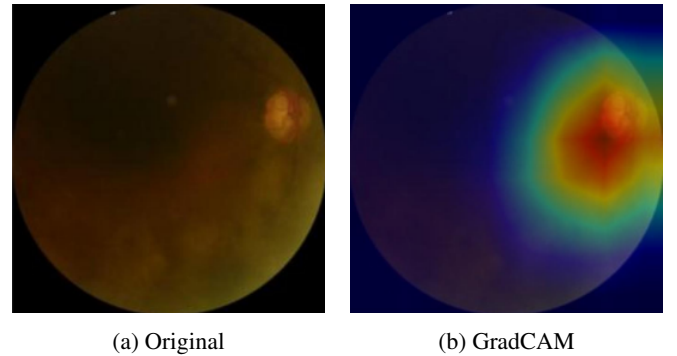
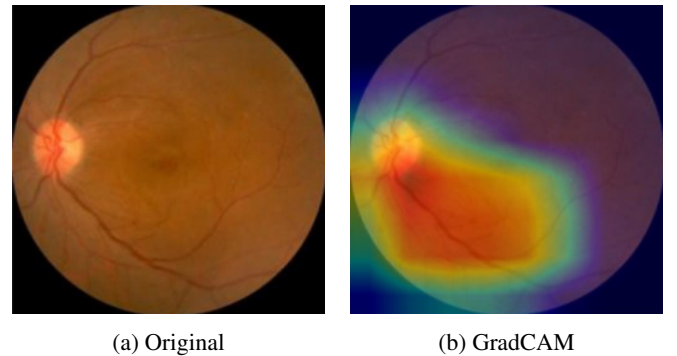
Table 4: Ensemble Comparison (%)

Method	Offsite	Onsite
Task 1-3 (Full FT)	78.99	81.84
Task 2-1 (Focal)	75.21	83.00
Task 2-2 (CB)	75.73	82.45
Simple Average	76.74	84.48
Weighted Average	76.29	84.48
Max Voting	75.45	81.91
Stacking	27.45	43.40

Stacking catastrophically fails (43.4%), likely due to: (1) insufficient validation data (200 samples) for meta-learner training, (2) overfitting to validation patterns that don’t generalize, (3) inability to learn meaningful combination rules with only 2-3 base models. This demonstrates that simpler ensemble methods are more robust for small medical datasets.

3.5. GradCAM Analysis

Figures 1, 2, 3 show attention maps validating clinically relevant feature learning.

**Fig. 1:** Glaucoma: Model focuses on optic disc cup-to-disc ratio (99.3% confidence)**Fig. 2:** DR: Attention on distributed hemorrhages and microaneurysms (99.5% confidence)

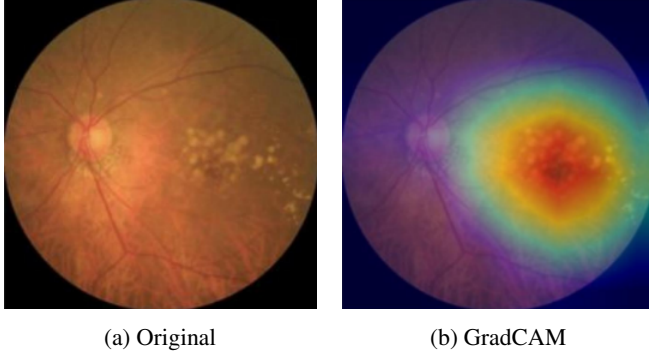


Fig. 3: AMD: Central macula attention for drusen detection (99.8% confidence)

4. DISCUSSION

4.1. Transfer Learning and Architecture

Full fine-tuning’s 27.73% relative improvement over no fine-tuning demonstrates that task-specific adaptation is crucial despite good pretrained features. ResNet18’s superior performance over EfficientNet (78.99% vs 71.96%) is noteworthy, as EfficientNet typically excels on larger datasets. This suggests ResNet’s simpler architecture with residual connections better handles limited training data, while EfficientNet’s compound-scaled architecture leads to overfitting.

4.2. Offsite vs Onsite Performance Gap

A critical observation is the divergence between offsite and onsite performance. Specialized techniques (Focal Loss, Class-Balanced Loss, attention mechanisms) often show negative or minimal impact on offsite validation (e.g., Focal: 75.21% vs BCE: 78.99%) but improve onsite performance (Focal: 83.00% vs BCE: 81.84%). This suggests: (1) offsite and onsite sets have different distributional characteristics, (2) techniques optimized for class imbalance generalize better to truly unseen data, (3) validation metrics alone are insufficient, held-out test evaluation is essential. This finding emphasizes the importance of the ensemble approach, which combines models with diverse validation characteristics.

4.3. Ensemble Strategy Analysis

Simple averaging’s success (84.48%) over more complex methods demonstrates that for small medical datasets, diversity matters more than learned combination rules. The near-identical weights [0.495, 0.505] from weighted averaging explain the identical performance with simple averaging. Each model contributes unique strengths: BCE (Task 1-3) excels on majority class DR (92.31% F1), while Focal and CB losses improve minority class detection (Glaucoma, AMD).

Stacking’s catastrophic failure (27.45% offsite, 43.40% onsite) provides an important lesson: meta-learning requires

sufficient training data. With only 200 validation samples and 2-3 base models, the meta-learner cannot learn robust combination rules and severely overfits. This contrasts with larger-scale ensemble applications where stacking typically excels.

4.4. Clinical Validation via GradCAM

GradCAM visualizations confirm anatomically appropriate attention patterns: optic disc for glaucoma (primary clinical indicator), distributed retinal regions for DR (systemic vascular disease), and central macula for AMD (drusen formation site). These patterns align with ophthalmological diagnostic criteria, validating that models learn medically meaningful features rather than dataset artifacts. The high confidence predictions ($\geq 99\%$) combined with correct anatomical focus support potential clinical utility, though rigorous clinical trials would be necessary before deployment.

4.5. Limitations and Future Work

Dataset size (800 images) limits generalization. Fixed threshold (0.5) may not be optimal per-disease, adaptive thresholding could improve performance. Future work: (1) Vision Transformers or Swin Transformers for better spatial modeling, (2) semi-supervised learning with unlabeled retinal images, (3) per-class threshold optimization, (4) larger ensembles with more diverse architectures.

4.6. Contribution Statement

Corentin Nicodème: Implemented all tasks, conducted experiments, performed analysis, and wrote the complete report.

5. CONCLUSION

We presented a comprehensive study on transfer learning for retinal disease classification, investigating fine-tuning strategies, specialized loss functions, attention mechanisms, and ensemble methods. Key findings: (1) ResNet18 outperforms EfficientNet on small datasets, (2) specialized techniques show better generalization to unseen data despite lower validation scores, (3) simple ensemble averaging significantly outperforms complex meta-learning on limited data. Our best ensemble achieves 84.48% F1-score (49.0% relative improvement over no fine-tuning baseline, 1.78% over best single model). GradCAM validates clinically relevant feature learning. These results provide practical guidelines for medical imaging with limited annotated data.

6. REFERENCES

- [1] M. Tan and Q. Le, “EfficientNet: Rethinking model scaling for CNNs,” *ICML*, 2019.
- [2] K. He *et al.*, “Deep residual learning for image recognition,” *CVPR*, 2016.
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A. COMPLETE EXPERIMENTAL RESULTS

Table 5: Offsite complete results (Precision / Recall / F1 by class) and Onsite score , EfficientNet and ResNet18

Arch.	Task	DR			Glaucoma			AMD			Avg F1	Onsite
		P	R	F1	P	R	F1	P	R	F1		
EfficientNet												
	Task 1.1	0.7459	0.6500	0.6947	0.5769	0.6122	0.5941	0.2464	0.7727	0.3736	0.5541	0.60382
	Task 1-2	0.8288	0.8643	0.8462	0.6923	0.5510	0.6136	0.7647	0.5909	0.6667	0.7088	0.74575
	Task 1-3	0.8540	0.8357	0.8448	0.7561	0.6327	0.6889	0.5769	0.6818	0.6250	0.7196	0.80772
	Task 2-1	0.8643	0.8643	0.8643	0.7250	0.5918	0.6517	0.6087	0.6364	0.6222	0.7127	0.81684
	Task 2-2	0.8529	0.8286	0.8406	0.7391	0.6939	0.7158	0.6522	0.6818	0.6667	0.7410	0.81362
	Task 3-1	0.8523	0.9071	0.8789	0.8235	0.5714	0.6747	0.7143	0.6818	0.6977	0.7504	0.81979
	Task 3-2	0.8779	0.8214	0.8487	0.7059	0.7347	0.7200	0.6500	0.5909	0.6190	0.7293	0.84211
ResNet18												
	Task 1.1	0.7172	0.5071	0.5941	0.5750	0.4694	0.5169	0.3019	0.7273	0.4267	0.5126	0.56668
	Task 1-2	0.8000	0.8857	0.8407	0.7368	0.2857	0.4118	0.7778	0.3182	0.4516	0.5680	0.61673
	Task 1-3	0.9041	0.9429	0.9231	0.8421	0.6531	0.7356	0.6957	0.7273	0.7111	0.7899	0.81839
	Task 2-1	0.8865	0.8929	0.8897	0.8250	0.6735	0.7416	0.5769	0.6818	0.6250	0.7521	0.82996
	Task 2-2	0.9262	0.8071	0.8626	0.7547	0.8163	0.7843	0.5769	0.6818	0.6250	0.7573	0.82450
	Task 3-1	0.9302	0.8571	0.8922	0.8571	0.7347	0.7912	0.5667	0.7727	0.6538	0.7791	0.82314
	Task 3-2	0.9298	0.7571	0.8346	0.7500	0.8571	0.8000	0.4167	0.6818	0.5172	0.7173	0.80596

B. ONSITE COMPARISON WITH REFERENCES

All trained models were submitted to the Kaggle competition (link) under username *Corentin Nicodème*.

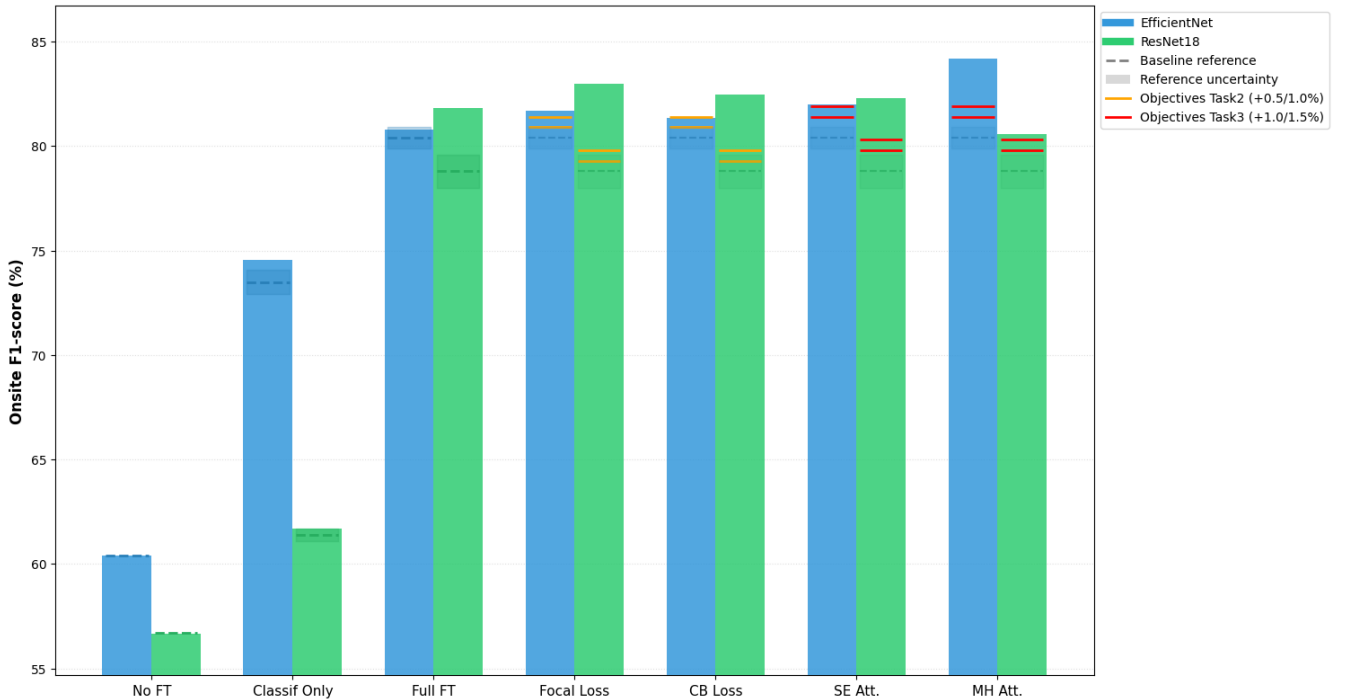


Fig. 4: Onsite comparative analysis: Trained models vs reference baselines & targets